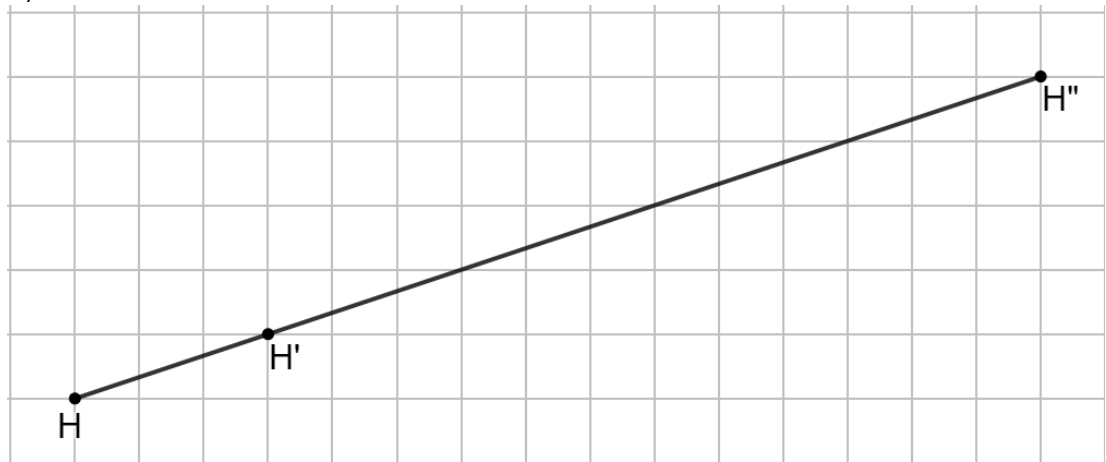


Points  $H$ ,  $H'$ , and  $H''$  are shown.



- 1. Determine the number of copies of  $\overline{HH'}$  that will fit on  $\overline{HH''}$ .
- 2. Use the definition of dilation to explain why  $\overline{HH''}$  is a dilation of  $\overline{HH'}$ .
- 3. Complete the statement.

Point 

$H$

$H'$

$H''$

 partitions 

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

 so that the ratio of  $\frac{HH'}{HH''}$  is 

$\frac{1}{4}$

$\frac{1}{5}$

Use the Pythagorean theorem to complete the table.

|    | Segment            | Length |
|----|--------------------|--------|
| 4. | $\overline{HH'}$   |        |
| 5. | $\overline{H'H''}$ |        |
| 6. | $\overline{HH''}$  |        |

7. Complete the statements.

Point  $H'$  is 

$\frac{1}{4}$

$\frac{1}{5}$

 of the distance from  $H$  to  $H''$ . The scale factor that can be

used to multiply 

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

 by to produce 

$\overline{HH'}$

$\overline{H'H''}$

$\overline{HH''}$

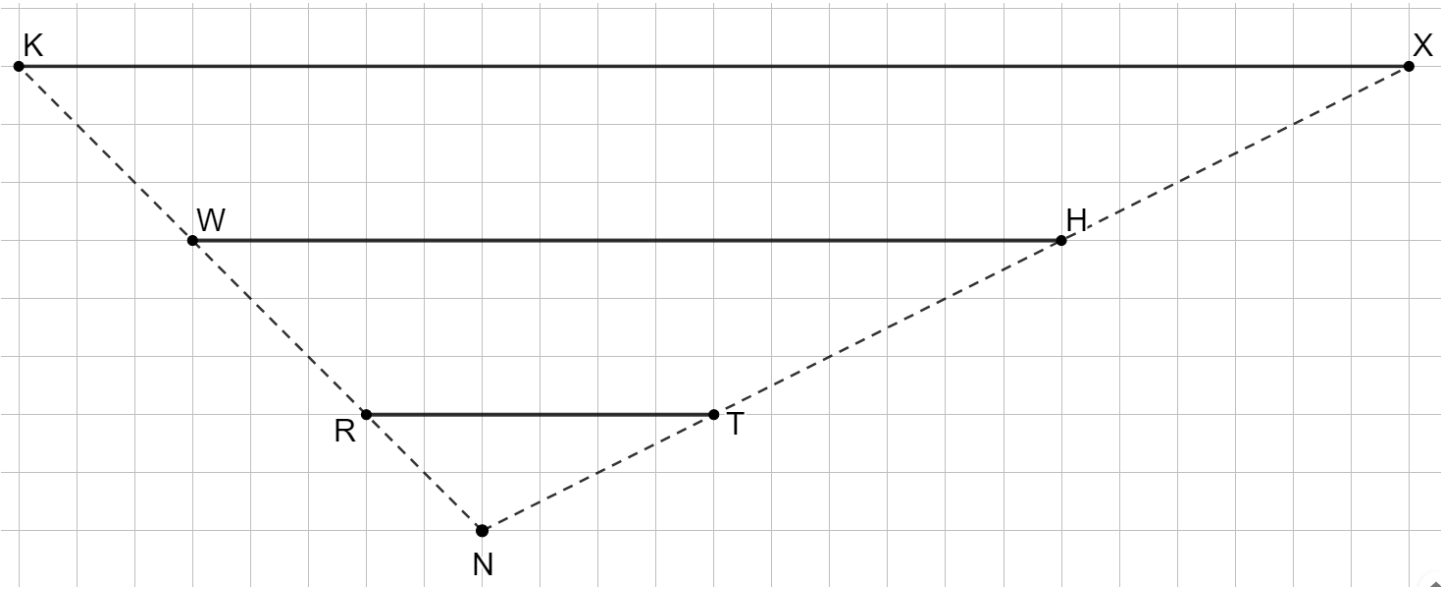
 is 

$\frac{1}{4}$

$\frac{1}{5}$

.

Line segments  $KX$ ,  $WH$ , and  $RT$  are shown where  $\overline{RT}$  is dilated using point  $N$  to create  $\overline{WH}$  and  $\overline{KX}$ .



Determine each of the ratios.

|     | Ratio   | Length |
|-----|---------|--------|
| 8.  | $NR:NW$ |        |
| 9.  | $RT:WH$ |        |
| 10. | $RT:KX$ |        |